

ACCENTURE × SUBAA · NOVACORP PEOPLE ANALYTICS CHALLENGE

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# What's driving the \$42M and where to act first

O for 4 · 15 August 2026

# You asked what's driving \$42M. We found you're counting \$14M of it.

Finance sized the annual people cost at **~\$42M** across three buckets: regrettable attrition (\$22–25M), disengagement (\$12–15M), hiring inefficiency (\$4–6M).

We did not set out to re-litigate that number. We set out to find which parts are **tractable**.

What we found first was a **measurement problem**, and it changes the shape of everything underneath it.

## WHERE WE'RE GOING

- 1 What your data is hiding
- 2 Where the loss actually sits
- 3 What it's worth if you act

*All of it uses your own numbers and Finance's own formulas.*

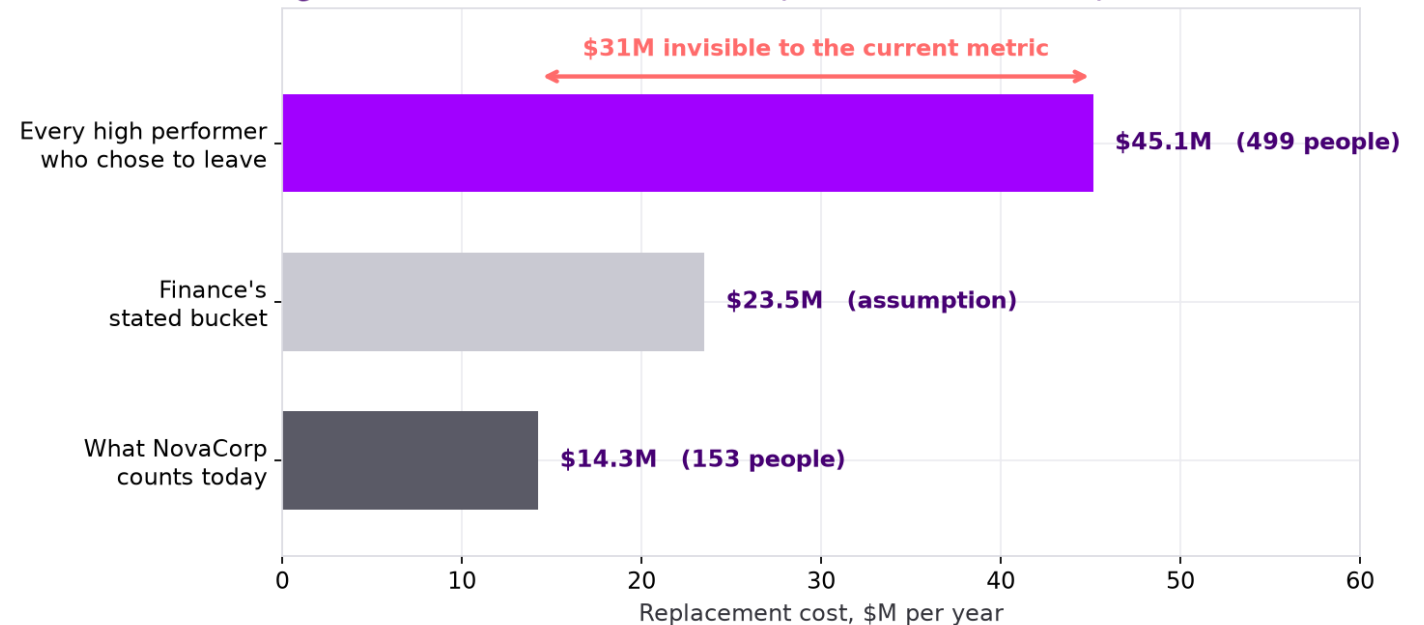
# You count \$14.3M of regrettable attrition. The real figure is \$45.1M.

HR's regrettable\_flag fires on **153 departures** over two years, which is **\$14.3M/yr.**

Apply a value-based definition, every High Performer or Outstanding employee who **chose** to leave, and it is **499 departures → \$45.1M/yr.**

**371 high-performing people** walked out and were never recorded as a loss.

Regrettable attrition: as counted, as Finance sized it, and as measured



The same replacement-cost formula Finance gave us, applied to a population their flag excludes.

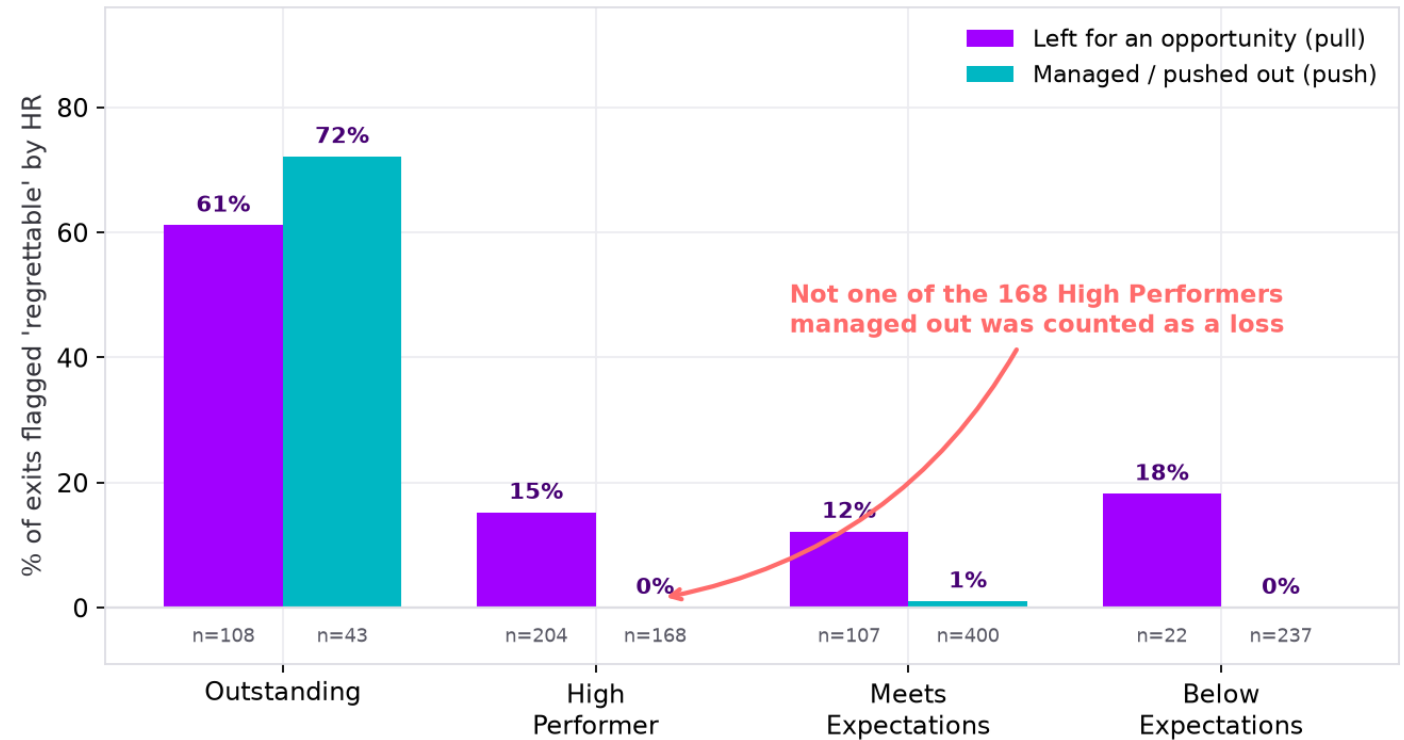
# The flag is a synonym for “Outstanding”, not a measure of value lost.

Flag rate by rating: **Outstanding 61–72%**. High Performer **15% (pull), 0% (push)**. Meets Expectations 12% / 1%.

Of the **168 High Performers who were managed out**, zero were flagged regrettable. Not a low rate. Zero.

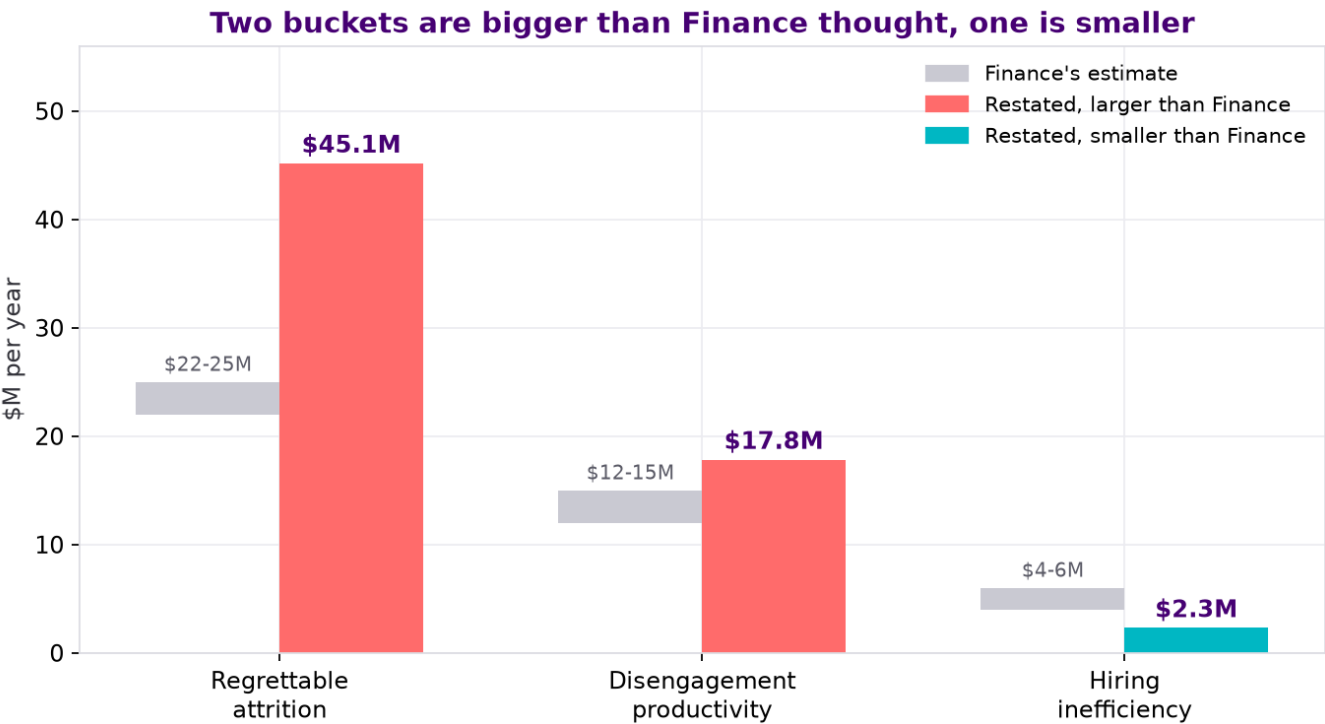
The flag is applied **after** the exit, by the same function that approved it. It records whether a departure was **convenient**, not whether it was **costly**.

Share of exits flagged regrettable, by performance band and pathway



Not bad faith. A retrospective, self-assessed metric that structurally exonerates the decision-maker is a justification, not a measurement.

It isn't \$42M. On your own formulas it's \$65M, and the shape is different too.



Finance sized the three buckets at \$38-46M a year, midpoint \$42M. On the data they total \$65.2M, about 1.6x that. All figures from cost\_model.py.

| Bucket                | Finance  | On the data |
|-----------------------|----------|-------------|
| Regrettable attrition | \$22-25M | \$45.1M     |
| Disengagement         | \$12-15M | \$17.8M     |
| Hiring inefficiency   | \$4-6M   | \$2.3M      |
| Total                 | \$38-46M | \$65.2M     |

Only one of these moves in your favour, and it's the smallest one. The two that moved against you were both sized with a definition that excluded the expensive cases.

## Entity\_B loses people at twice Entity\_A's rate, and it's the one still on its own HR system.

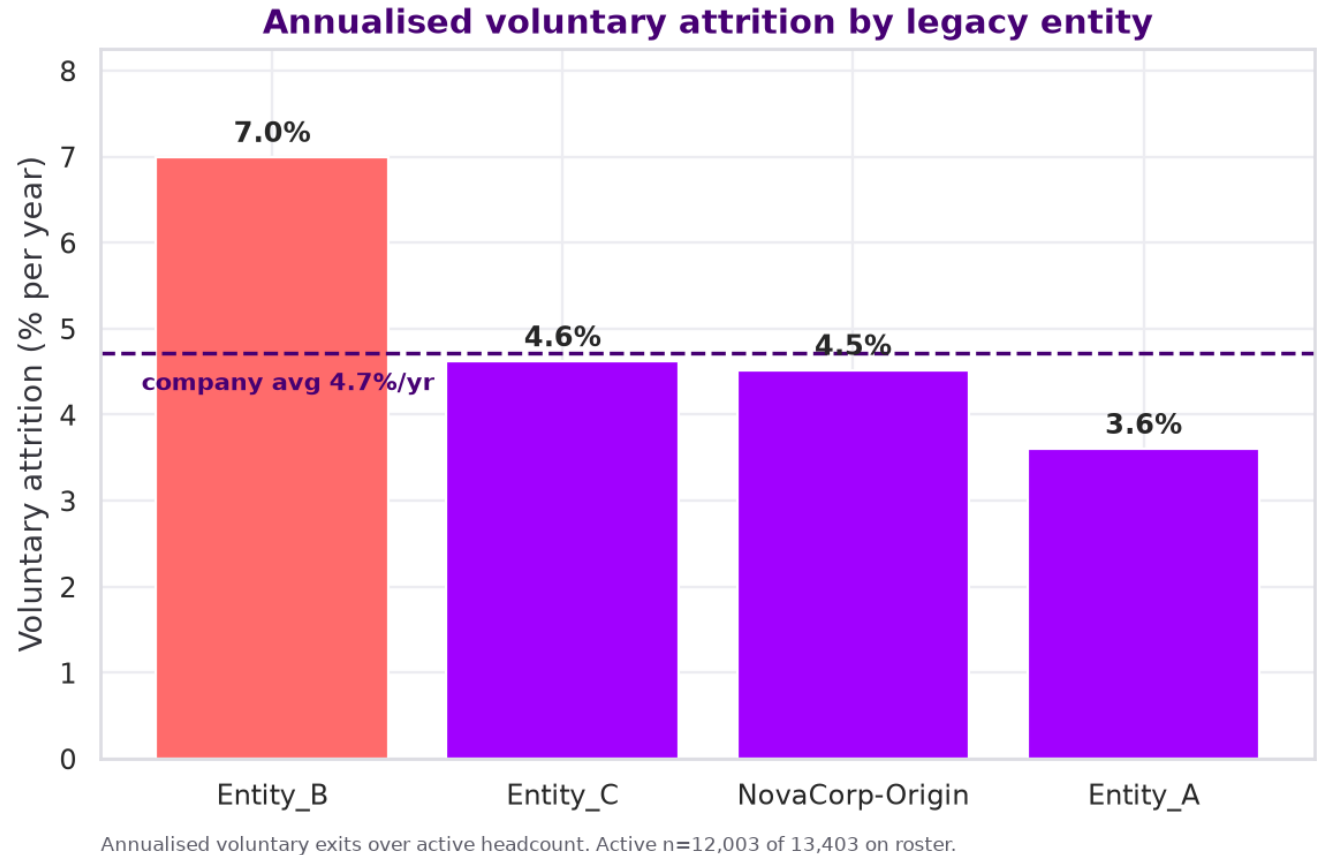
NovaCorp-Origin **4.5%** · Entity\_A **3.6%** · **Entity\_B 7.0%** · Entity\_C **4.6%**, per year.

Entity\_B vs Entity\_A on 1,601 and 1,804 active staff. The most statistically solid finding in the analysis.

Entity\_B was acquired in FY2023 and **still runs on BambooHR**, two years on. Entity\_A was absorbed onto the core system and is now the healthiest cohort in the company.

**It isn't department mix.** Controlling for department, role level, pay and potential, Entity\_B still leaves at **1.95×** Entity\_A (appendix A15).

Entity\_B isn't where the most dollars sit. NovaCorp-Origin holds \$29.2M of the \$45.1M on headcount alone. Entity\_B is where the leverage is: \$8.4M in 1,601 people you already know how to fix.



## Six of eight engagement dimensions are fine. Two collapsed.

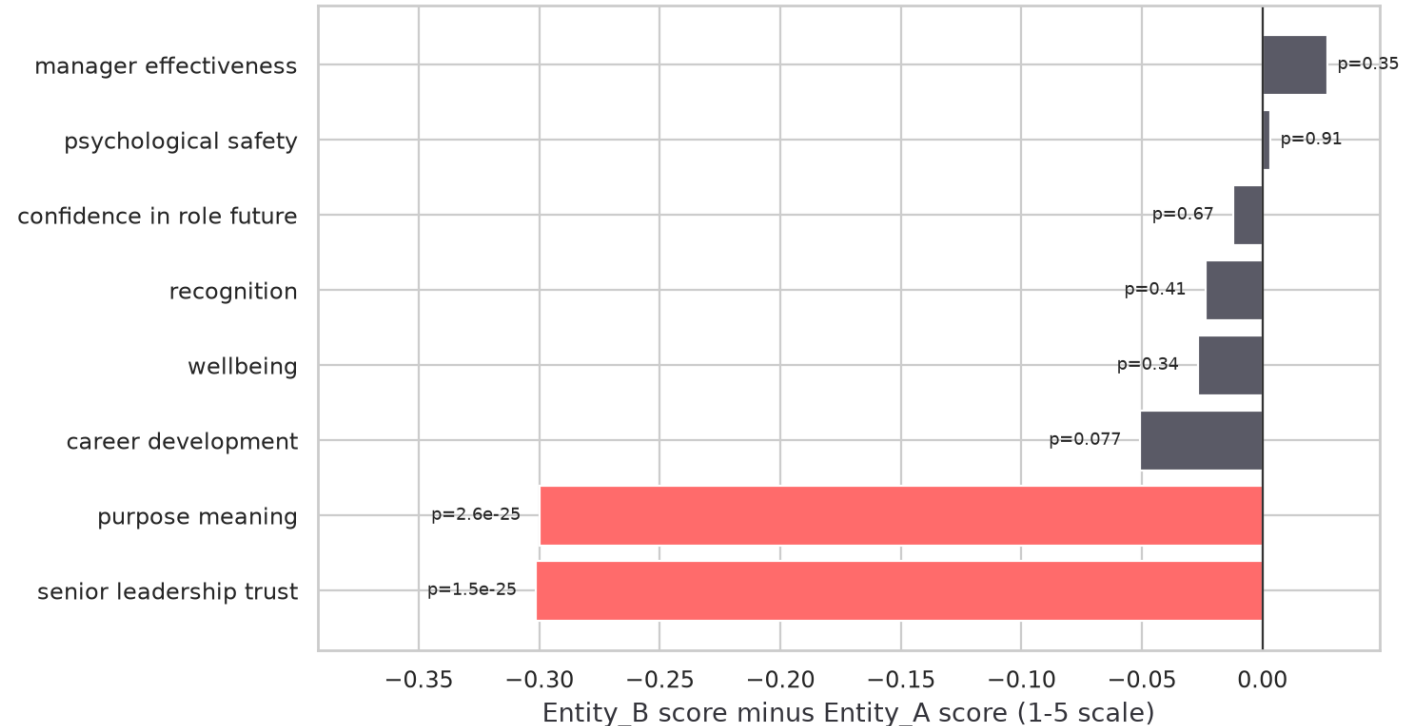
**Manager effectiveness:** 3.344 vs 3.371, **no difference** ( $p = 0.35$ ).

**Psychological safety:** 3.351 vs 3.354, **no difference** ( $p = 0.91$ ).

**Senior leadership trust**  $-0.301$  ( $p = 1.5 \times 10^{-25}$ ). **Purpose & meaning**  $-0.300$  ( $p = 2.6 \times 10^{-25}$ ).

And it isn't money. Entity\_B sits at **0.961** compa-ratio against NovaCorp-Origin's **0.937**. They are paid better and leave anyway.

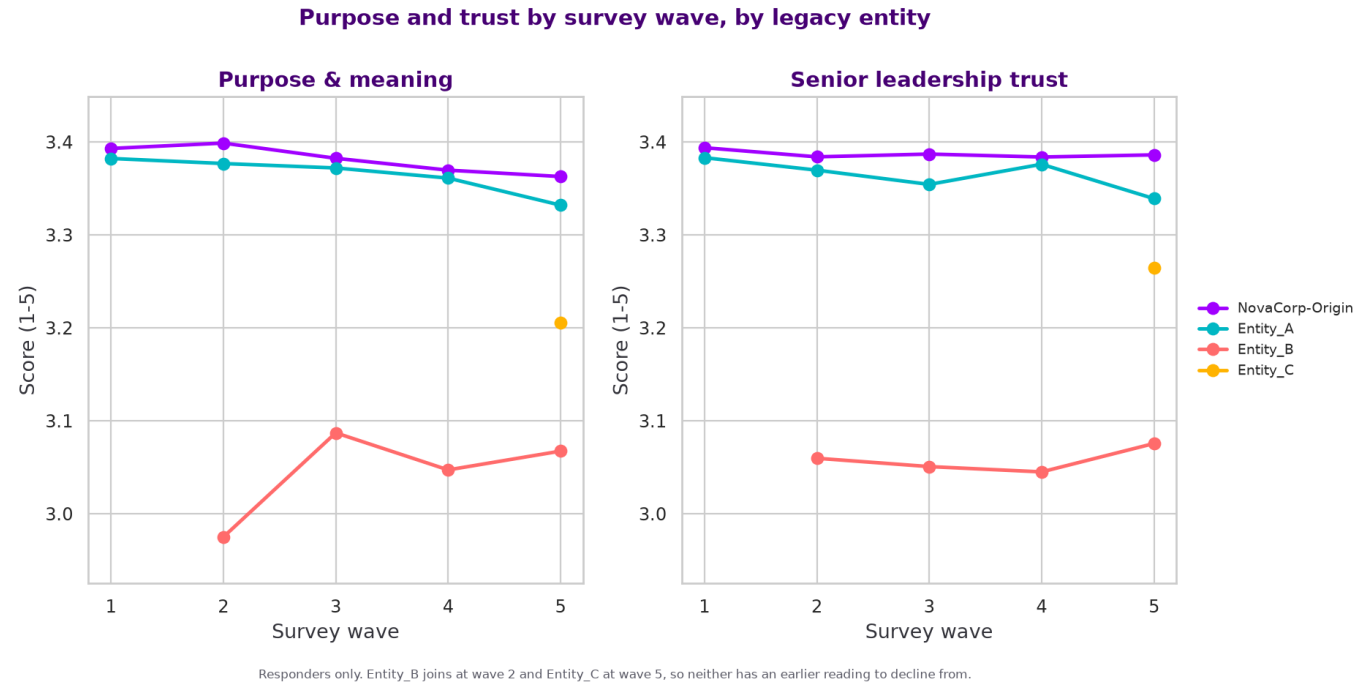
Entity\_B minus Entity\_A, all eight engagement dimensions



Employee-level means. Entity\_A n=1,938, Entity\_B n=1,697. Welch t-test, coral where  $p < 0.01$  and gap  $> 0.1$ pt.

Why it never showed on your dashboard: the composite index averages all eight dimensions, so two collapsed scores are diluted by six healthy ones. A gap of 0.085 that looks like nothing.

## There was no decline to detect. Entity\_B's first-ever survey was already broken.



Entity\_B's **first** measurement: purpose 2.975, trust 3.060, against NovaCorp-Origin's steady 3.38. A year later: 3.067. **Flat.**

Survey response rate: Entity\_B **62.6%** vs Entity\_A **83.8%**, 21 points lower.

**307 employees were never issued a survey at all.** 152 of the mid-window leavers among them sit on the Entity\_B and Entity\_C legacy systems.

They weren't silent. **They were never asked.**



# Entity\_A is at 3.6%, below NovaCorp's own 4.5%. Integration works.

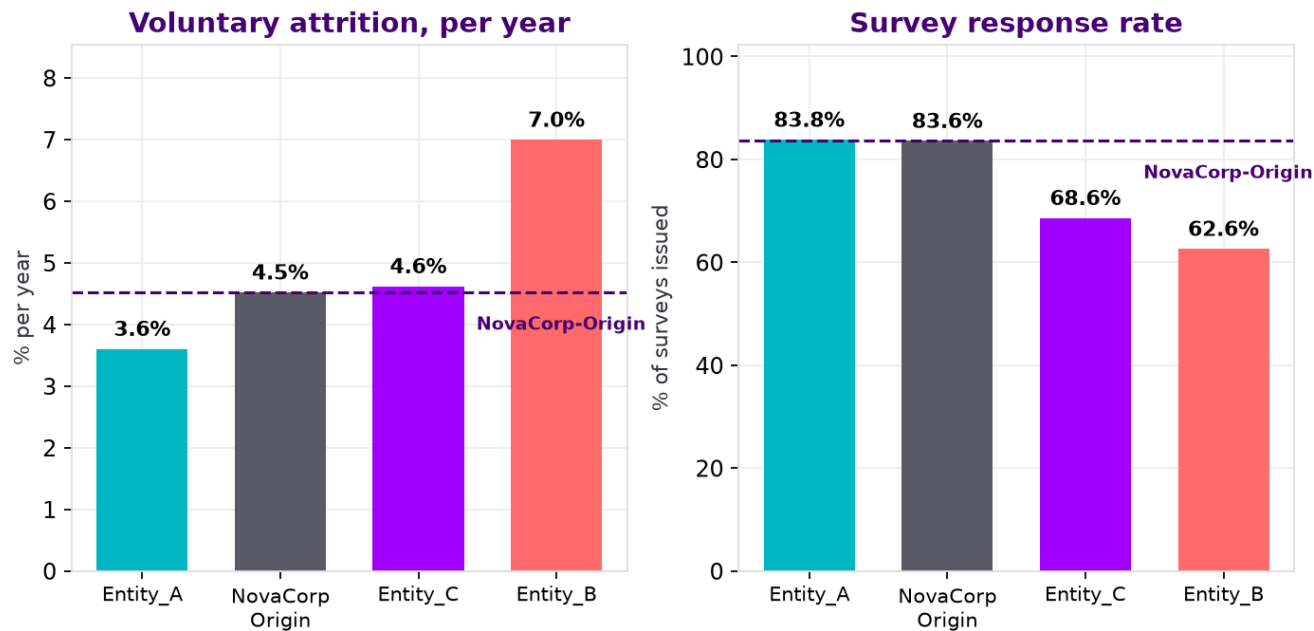
Entity\_A (FY2022, fully integrated) is now the **safest cohort in the company**, better than NovaCorp-Origin itself.

Its survey response rate has recovered to **83.8%**, indistinguishable from NovaCorp-Origin.

Entity\_C (late FY2024) is at **4.6%** and statistically indistinguishable from Entity\_A. It is **not** a second Entity\_B yet.

*This turns the recommendation from “we hope this helps” into “you have already done this once, in this company, with this workforce.” It also sets your success measure.*

Attrition and survey response, by legacy entity



Attrition is annualised voluntary exits over active headcount (n=12,003). Response rate is the mean of each employee's own rate across five waves (n=13,096 employees).

Attrition is annualised voluntary on active headcount. Response rate is the mean of each employee's own rate across five waves. Entity\_A vs Entity\_C,  $p = 0.101$ , not significant.

# Three actions, in order. The first one is free.

## 1 Redefine what counts as a regrettable loss

*~\$0 • policy • this quarter*

Include High Performer alongside Outstanding. Separate the flag from the function that approved the exit. Review quarterly. Restates your baseline immediately and fixes next year's \$42M at source.

## 2 Finish the Entity\_B integration, as a trust problem not a systems problem

*inside the guided \$40-50M FY26 integration budget*

Migrate off BambooHR. Stabilise the Senior Manager layer first — Entity\_B loses L3s at 9.8%/yr against Entity\_A's 2.9%, a 3.4× gap — or there is nobody credible left to communicate the integration. Then leadership visibility and honest communication. Not manager training.

## 3 Measure acquired cohorts from day one, at cohort level

*~\$0*

Baseline every acquired group against company norms at their first survey. Track response rate as a health metric in its own right. Fix the 307 who were never surveyed.

*Notice what's not on this list: a pay round, a manager training programme, and cutting agency recruitment. The data rules out all three.*

\$71M addressable. The cheapest lever is the largest.

| Lever                                        | Addressable | @20%   | @40%    | Cost to act         |
|----------------------------------------------|-------------|--------|---------|---------------------|
| Redefine regrettable_flag + quarterly review | \$45.1M     | \$9.0M | \$18.1M | ~\$0 (policy)       |
| Disengagement (813 staff below 2.5)          | \$17.8M     | \$3.6M | \$7.1M  | programme-dependent |
| Early-tenure / acquisition onboarding        | \$7.9M      | \$1.6M | \$3.2M  | ~\$500–800/head     |
| of which Entity_B specifically               | \$6.2M      | \$1.2M | \$2.5M  |                     |

The 20% and 40% are assumptions, not findings. That's the band typically claimed for targeted retention programmes. We show both so the decision doesn't depend on the optimistic one. The first lever is a definition change, so its return isn't capped by budget at all.

\$7.9M is the corrected figure.

An earlier draft said \$29.6M. That counted the baseline early attrition every employer carries, not the excess attributable to the acquisition cohorts. Method and restatement in appendix A7.

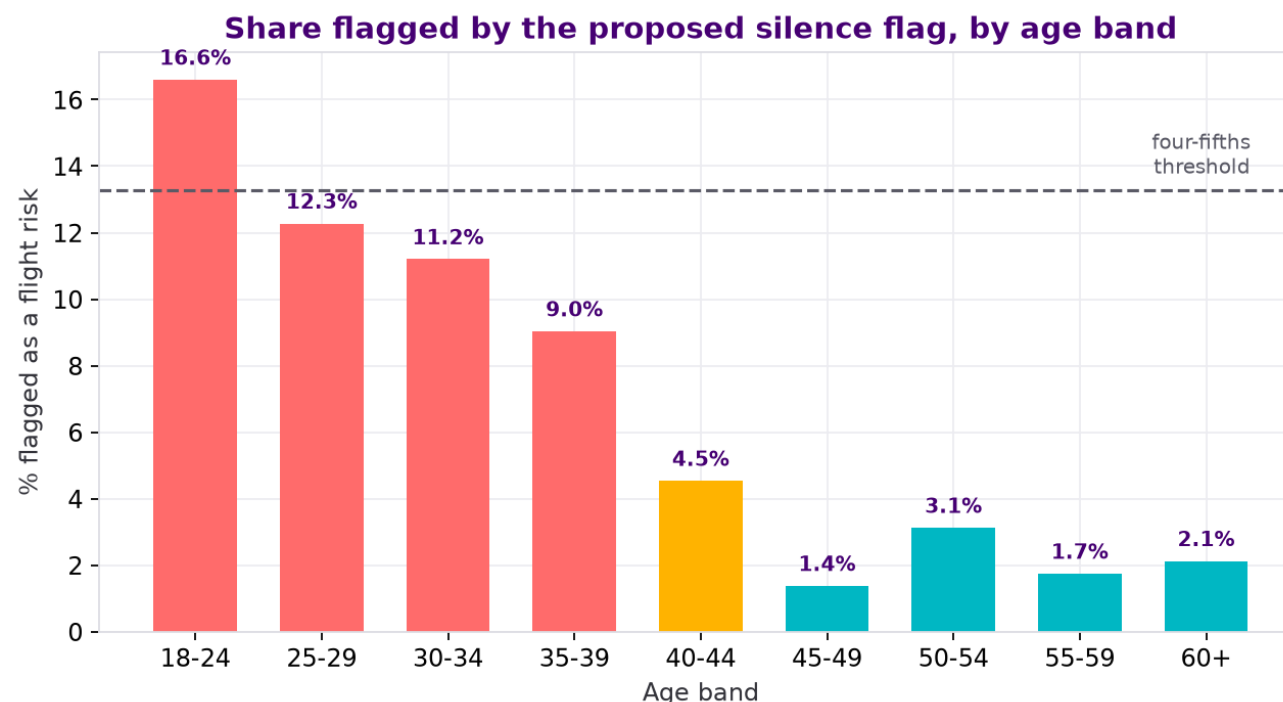
## A “who stopped answering” flag looks free. It flags 16.6% of 18–24s and 1.4% of 45–49s.

The obvious move from our own analysis is a flight-risk score built on survey silence. **We recommend against deploying it at individual level.**

Under the **four-fifths rule** it fails on age (ratio **0.08**, a twelvefold gap), on role level (L1 9.2% vs L5 1.4%) and on acquisition cohort (Entity\_C 31.7% vs NovaCorp-Origin 2.4%).

It is also **wrong most of the time: 901 false positives** against 263 true ones, **precision 23%**. The dominant outcome is a career conversation about someone who was never leaving.

Low response is concentrated in the acquisition cohorts, so the flag is substantially measuring **integration failure**, not individual intent.



*We're showing you the thing we decided not to recommend. In a performance or redundancy conversation this is a discrimination exposure that costs more than the attrition does.*

# Team-level diagnostic, disclosed to staff, never an individual score.

- 1 Never as an individual score handed to a manager.**  
23% precision. Three in four flagged people were never leaving.
- 2 Team level only, minimum 8 responses.**  
Reported as a diagnostic, not a prediction.
- 3 Tell employees the participation metric exists.**  
If the survey was sold as confidential, quietly re-using non-response as a personal risk score breaks that promise. Once staff work it out, response rates collapse and you lose the instrument entirely.

## WHAT WE DELIBERATELY DID NOT DO

### **No demographic features in any model.**

Gender, age and cultural background appear only as fairness-audit dimensions, never as predictors.

### **No individual employees or managers named.**

Manager results aggregated, suppressed below 8 reports.

### **No causal language.**

Association only. We never claim one thing caused another.

# The limits of this analysis, stated before you ask.

**Two-year window, annualised.** All rates are  $\text{voluntary exits} \div 2 \text{ years} \div \text{active headcount} = 4.7\%/yr$ . Your FY2025 Annual Report's "10.4% voluntary attrition" is total attrition including 267 involuntary exits across the full two years. It reproduces exactly as  $1,400/13,403$ . **We restate; we don't accuse.**

**We cannot see pre-acquisition Entity\_B.** They may have arrived unhappy from their previous employer. We genuinely cannot rule this out.

**Engagement is only observed for responders**, and leavers responded less, so every disengagement figure is a lower bound.

**Association, not causation.** Senior Manager churn and collapsed trust move together. We cannot say which drives which.

**regrettable\_flag, performance\_band\_at\_exit and stated\_exit\_reason are retrospective HR judgements.** We use them as the object of analysis, not as ground truth.

**The 2025-H2 review cycle is missing**, so performance data is up to 12 months stale for some leavers.

**Small samples above Level 4.** Nothing above L4 is reliable: 279 people at L4, 103 above it.

*On a strict voluntary reading you have already met the FY2026 sub-9.5% target. That's worth resolving before the next board pack.*

## Start with the free one.

### THIS QUARTER

#### Redefine regrettable.

Cost ~\$0. It restates your baseline and stops you funding a budget line built on a blind spot.

### THIS HALF

#### Finish Entity\_B.

Senior Manager layer first, then leadership visibility.  
\$8.4M concentrated, inside a budget you have already guided.

### ONGOING

#### Baseline every acquired cohort from day one.

Entity\_C is healthy today. It does not have to become the next Entity\_B.

#### You already have the money set aside.

The Annual Report commits to a People Reinvention Programme three times. This is what we'd spend it on, and this is what we'd measure.

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# Appendix

A1 to A14. Method, assumptions, every test, and where each number comes from.



# Constants

Every constant below is published by Finance in the case brief §6. Nothing outside this table is used as a benchmark.

| Constant                        | Value                     | Used for                      |
|---------------------------------|---------------------------|-------------------------------|
| Replacement cost multiplier     | 1.50 × annual base salary | attrition buckets             |
| Backfill rate                   | 85% of vacated roles      | attrition buckets             |
| Superannuation on-cost          | 12.0% of base             | all salary figures grossed up |
| Disengagement productivity loss | 15% of base salary/yr     | disengagement bucket          |
| Agency fee rate                 | 18% of first-year base    | hiring bucket                 |
| Direct hire benchmark           | \$5,500 fully loaded      | hiring bucket                 |

The only numbers we introduce beyond this table are the 20% and 40% intervention-effectiveness rates on slide 11. Those are labelled as assumptions on the slide itself and shown at both ends, so no conclusion depends on the optimistic one.

# Population decisions

The brief asks explicitly that these be documented. Each choice is stated with the reason it was made.

| Decision                  | Choice                                             | Why                                                                         |
|---------------------------|----------------------------------------------------|-----------------------------------------------------------------------------|
| Observation window        | 1 Jan 2024 – 31 Dec 2025 = 2.0 years               | Stated in brief §4. Every rate divided by 2 to annualise.                   |
| Attrition denominator     | Active headcount (12,003)                          | A rate quoted against a roster that still contains leavers understates it.  |
| Which exits count as cost | Voluntary only (1,133 of 1,400)                    | An involuntary exit is a decision the company made, not a loss it suffered. |
| “High value”              | Band at exit $\in$ {Outstanding, High Performer}   | The only value proxy available at the point of exit.                        |
| Disengaged                | Index < 2.5 over the 2 most recent responded waves | “Persistently”, per the brief. Single-wave dips are noise.                  |
| Non-responders            | Retained, not dropped                              | Brief §4 warns these rows are intentional. They carry the silence signal.   |
| Early-tenure population   | Hired on/after 1 Jan 2024 only                     | Removes the left-truncation bias documented in A6 §1.1.                     |

## Cost formulas

Three formulas produce every dollar figure in the deck. All are the brief's own, applied to the populations defined in A2.

### Replacement cost

$$1.50 \times 0.85 \times \Sigma(\text{salary\_at\_exit} \times 1.12) \div 2.0$$

Applied to voluntary exits rated Outstanding or High Performer. Gives \$45.1M/yr.

### Disengagement

$$\Sigma(\text{salary\_active} \times 1.12) \times 0.15$$

Applied to the 813 active staff whose index is below 2.5. Gives \$17.8M/yr.

### Agency premium

$$\Sigma \max(\text{salary} \times 0.18 - 5,500, 0) \div 2.0$$

Applied to the 274 agency hires made inside the window. Gives \$2.3M/yr.

The early-tenure lever (\$7.9M) uses the replacement-cost formula applied to the excess exits over a 6.4% NovaCorp-Origin new-joiner baseline, not to all early exits. Method in A7.

# Fairness method and results

Four-fifths (80%) rule, the standard EEOC adverse-impact screen: a group's rate divided by the highest group's rate. Below 0.80 warrants investigation.

| Flag tested           | Dimension                          | Worst ratio | Detail                                          | Verdict       |
|-----------------------|------------------------------------|-------------|-------------------------------------------------|---------------|
| Proposed silence flag | Age band                           | 0.08        | 18-24: 16.6% vs 45-49: 1.4% (n = 1,253 / 1,017) | Fails badly   |
| Proposed silence flag | Role level                         | 0.15        | L1 9.2% vs L5 1.4% (n = 6,931 / 74)             | Fails         |
| Proposed silence flag | Acquisition cohort                 | 0.08        | Entity_C 31.7% vs NovaCorp-Origin 2.4%          | Fails         |
| HR regrettable_flag   | Gender, within Outstanding leavers | 0.60        | M 74.6% / F 59.4% / NB 44.4% (n = 67 / 64 / 9)  | Under-powered |

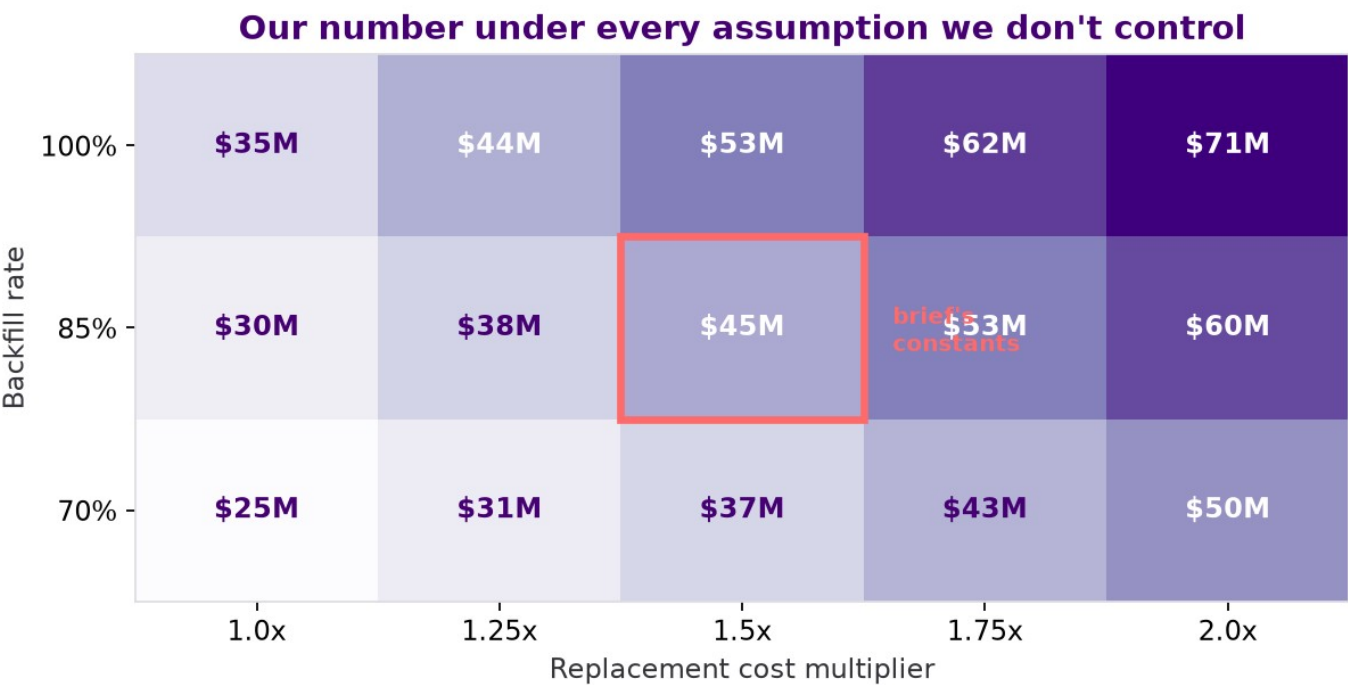
**The silence flag result is unambiguous.** It fails on three dimensions at once, on groups of 900 to 1,900 people. This is why we recommend against individual-level deployment on slide 12.

**The regrettable\_flag result is suggestive but under-powered.** With 9 non-binary staff in the comparison we report it as a governance risk requiring audit, not as established discrimination. Saying more than the sample supports would be the same error we are criticising.

All figures reproduce from ethics\_audit.py. Groups below n = 20 (flag test) and n = 50 (silence test) are suppressed rather than reported.

# Sensitivity

How the \$45.1M headline moves across the two assumptions we do not control.



Full range across the grid: **\$24.8M to \$70.8M**.

We quote **\$34-56M** as the defensible band, roughly  $\pm 25\%$  around the brief's own constants.

**Every cell in the grid exceeds the \$14.3M currently counted.**  
The finding does not depend on the assumption. That is the point of showing the grid rather than a single number.

Replacement multiplier 1.0-2.0x, backfill 70-100%. Brief's constants are 1.50x and 85%.

# Data quality register — issues found

The brief calls data preparation “a substantive analytical step, not a preprocessing formality”. This records every issue, and what we did about it.

| #   | Issue                                                                                                                                               | What we did                                                                                                                                          |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.1 | hire_date is a system-entry date for acquired staff. Entity_C's whole roster reads as under 9 months tenure. Material.                              | Restricted all early-tenure analysis to in-window hires, so everyone is observed from tenure 0. Changed the lever from \$29.6M to \$7.9M. See A7.    |
| 1.2 | The 2025-H2 performance cycle is absent, so band at exit can be up to 12 months stale.                                                              | Used performance band as a value proxy at exit, not a current-state measure, and flagged staleness wherever it carries a dollar.                     |
| 1.3 | 307 employees have no engagement records. 171 were employed while surveys ran and were still never asked; 152 sit on Entity_B and Entity_C systems. | Excluded from the silence-flag population, because “never issued” is not “issued and declined”. Reported separately as a data-fragmentation finding. |
| 1.4 | 10,264 rows have response_flag = False with null dimensions. Intentional, per the brief.                                                            | Retained every row. Response rate used as a variable in its own right. Scores averaged per employee before any group comparison.                     |
| 1.5 | “Push” does not mean disengagement-driven. 267 of 955 push exits are involuntary. Definitional.                                                     | Removed any claim that 68% push implies 68% preventable. All cost figures use voluntary exits only.                                                  |
| 1.6 | Small samples above Level 4: L5 78, L6 16, L7 8, L8 1.                                                                                              | Report nothing above L4 as a finding. Where a senior gap appears, state the n and decline to conclude.                                               |
| 1.7 | The Annual Report's “voluntary attrition” is total attrition over two years on the full roster.                                                     | Restated as annualised voluntary on active (4.7%/yr) and footnoted the denominator on every slide carrying a rate.                                   |

# Data quality register — checks that came back clean

Documenting what we verified and found sound matters as much as what we found broken.

| Check                                              | Result                                 |
|----------------------------------------------------|----------------------------------------|
| Duplicate employee_id in employees.csv             | 0                                      |
| attrition_log rows with no matching employee       | 0                                      |
| Employees marked departed with no attrition record | 0 — exact 1:1 on all 1,400             |
| exit_date disagreement between the two files       | 0 of 1,400                             |
| salary vs salary_at_exit disagreement              | 0 of 1,400                             |
| Date formatting variation                          | None — all YYYY-MM-DD across all files |
| Negative or impossible tenure values               | 0                                      |
| Survey rows dated after an employee's exit date    | 0 — the link is not an artifact        |
| Nulls outside the two expected fields              | None                                   |

WHAT THIS MEANS

**The data is materially cleaner than the brief's framing implies.**

The substantive issues are semantic, not structural. Fields that are well-formed but do not mean what their names suggest: hire\_date, pathway, and the Annual Report's “voluntary attrition”.

That is the harder failure mode to catch, and it is where our data-preparation effort went.

# Early-tenure correction: \$29.6M → \$7.9M

The largest single correction we made to our own work. Reproduces with `cost_model.py` §3 and §5, cross-checked independently by `cost_fix2.py`.

## WHAT WAS WRONG

The original defined an early exit as anyone departed with tenure  $\leq 12$  months, measured against **all** hires of that source.

But 94.7% of NovaCorp-Origin staff were hired before the window opened, so we never see their first year. Entity\_C cannot contain anyone above 9 months' tenure, so **100% of its exits are early exits by construction**.

The comparison measured **who we can observe**, not who leaves early.

## THE FIX

| Cohort          | n     | early exits | rate  |
|-----------------|-------|-------------|-------|
| NovaCorp-Origin | 455   | 29          | 6.4%  |
| Entity_C        | 1,014 | 85          | 8.4%  |
| Entity_B        | 1,884 | 192         | 10.2% |

Restricted to people hired inside the window, so everyone is observed from tenure 0. Cross-checked with a left-truncated Kaplan-Meier.

| Cohort   | rate vs 6.4% baseline | excess exits | cost      |
|----------|-----------------------|--------------|-----------|
| Entity_B | 10.2%                 | 72           | \$6.2M/yr |
| Entity_C | 8.4%                  | 20           | \$1.8M/yr |
| Total    |                       | 92           | \$7.9M/yr |

Only the excess over baseline is addressable. The old figure counted the ~6.4% early attrition every employer carries. Two claims were retracted with it: see A12.



## Statistical test register — findings we rely on

Every comparison in this deck, the test behind it, and the result. Run with `scipy.stats`, threshold  $p < 0.05$ . Survey comparisons average each employee across waves first, so nobody is counted five times. Attrition rates are annualised voluntary on active headcount.

| #  | Claim                                         | Test                       | Result                        | p                              | n             |
|----|-----------------------------------------------|----------------------------|-------------------------------|--------------------------------|---------------|
| 1  | Attrition differs by legacy entity            | Chi-square, 4 groups       | 4.5 / 3.6 / 7.0 / 4.6 %/yr    | $6.7 \times 10^{-9}$           | 12,003        |
| 2  | Entity_B vs Entity_A specifically             | Chi-square, pairwise       | 7.0% vs 3.6%/yr               | $8.0 \times 10^{-9}$           | 1,601 / 1,804 |
| 3  | Senior leadership trust collapsed             | Welch's t                  | 3.352 → 3.051 (−0.301)        | $1.5 \times 10^{-25}$          | 1,938 / 1,697 |
| 4  | Purpose & meaning collapsed                   | Welch's t                  | 3.358 → 3.058 (−0.300)        | $2.6 \times 10^{-25}$          | 1,938 / 1,697 |
| 5  | ...and holds among active staff only          | Welch's t                  | same direction                | $4.6 \times 10^{-25}$          | active only   |
| 6  | Manager effectiveness is NOT different        | Welch's t                  | 3.344 vs 3.371                | 0.35 — null                    | 1,938 / 1,697 |
| 7  | Psychological safety is NOT different         | Welch's t                  | 3.351 vs 3.354                | 0.91 — null                    | 1,938 / 1,697 |
| 8  | Entity_B response rate is lower               | One-way ANOVA              | 83.6 / 83.8 / 62.6 / 68.6 %   | $7.0 \times 10^{-304}$         | 13,096        |
| 9  | Entity_B is paid above NovaCorp-Origin        | One-way ANOVA              | 0.937 / 0.959 / 0.961 / 0.962 | $2.2 \times 10^{-71}$          | 13,403        |
| 10 | Entity_B loses Senior Managers (L3)           | Chi-square                 | 2.9% vs 9.8%/yr               | 0.0047                         | 140 / 123     |
| 11 | HIPO staff leave at 1.6×                      | Chi-square                 | 7.3% vs 4.5%/yr               | $1.2 \times 10^{-7}$           | 12,003        |
| 12 | Attrition NOT concentrated under bad managers | Chi-square goodness-of-fit | worst 10% hold 18.4% of exits | 0.9957 — null                  | 1,196 mgrs    |
| 13 | Regrettable flag ≠ value-based definition     | Chi-square                 | 97 of 312 overlap             | $< 10^{-6}$                    | 1,400         |
| 14 | Entity_B new joiners leave early more         | Two-proportion z           | 10.2% vs 6.4%                 | 0.012 (z=2.50)                 | 1,884 / 455   |
| 15 | Composite engagement barely differs           | One-way ANOVA              | 3.377 / 3.366 / 3.280 / 3.346 | $2.5 \times 10^{-9}$ — trivial | 13,096        |

Row 15 is deliberately included. A significant p-value on a 0.085-point gap is the clearest illustration of why significance is not importance.

# Statistical test register — the nulls that matter

Findings that came back null. Each one rules out an expensive intervention, which makes them as load-bearing as the positives.

| Claim tested                              | Test           | Result                      | p                     | Why the null matters                                                        |
|-------------------------------------------|----------------|-----------------------------|-----------------------|-----------------------------------------------------------------------------|
| Manager effectiveness differs in Entity_B | Welch's t      | 3.344 vs 3.371              | 0.35                  | Rules out a manager training programme, the most expensive obvious response |
| Psychological safety differs in Entity_B  | Welch's t      | 3.351 vs 3.354              | 0.91                  | Rules out a team-climate intervention                                       |
| Entity_B is underpaid                     | One-way ANOVA  | 0.961 vs 0.937 — paid above | $2.2 \times 10^{-71}$ | Rules out a retention pay round. They are paid better and leave anyway      |
| Attrition concentrates under bad managers | Chi-square GoF | worst 10% hold 18.4%        | 0.9957                | Rules out performance-managing a manager tail                               |
| Purpose separates leavers within Entity_B | Welch's t      | 2.929 vs 3.070              | 0.084                 | Rules out individual targeting on purpose scores                            |
| Entity_A vs Entity_C attrition            | Chi-square     | 3.6% vs 4.6%/yr             | 0.101                 | Entity_C is not a second Entity_B — yet                                     |

Three of the six nulls each rule out a seven-figure programme. Slide 10's “notice what's not on this list” rests entirely on this page.

# Glossary

*No statistics background assumed. Every term used in this deck, in plain English.*

## p-value

If there were no real difference between two groups, how surprising would a gap this big be from luck alone? Below 0.05 by convention. It is not the size of an effect.

## Statistically significant

Shorthand for “unlikely to be a coincidence”. It does not mean big or important.

## Chi-square test

Used when comparing rates or categories between groups.

## Welch's t-test

Used when comparing the average of a number between two groups.

## Left truncation

When people only become visible partway through their tenure. Ignoring it makes newer cohorts look like they leave faster. See A6 §1.1.

## Pseudo-replication

Counting the same person once per survey wave, which makes results look more certain than they are. Avoided here.

## Four-fifths rule

The standard EEOC adverse-impact screen. Below 0.80 warrants investigation.

## Precision / recall

Of everyone a flag identifies, precision is the share who actually leave. Recall is the share of leavers it catches.

## Compa-ratio

Salary divided by the midpoint of the pay range for that role and level. 1.00 = paid exactly at midpoint.

## HIPO

“High potential.” A talent-review tag marking a likely future leader. Separate from a performance rating.

## Push vs pull

Push = the organisation drove or managed the exit. Pull = an outside opportunity drew them away. Push includes genuinely involuntary exits and does not mean disengagement-driven.

## Regrettable attrition

HR's label for a departure the company didn't want. Here it is a retrospective judgement recorded after the exit, not a measurement. That is the subject of slides 3 and 4.

## Composite index

One number made by averaging several scores. It can hide a serious problem in one component by averaging it against healthy ones, which is what happened to Entity\_B.

## Backfill rate

The share of vacated roles actually refilled, 85% per the brief. Unfilled roles incur no replacement cost.

## FAR

Financial Accountability Regime. Australian legislation effective March 2024 imposing personal accountability on senior financial executives. Named in NovaCorp's Annual Report as a driver of Risk & Compliance attrition.

# Annual Report reconciliation

We cross-checked against NovaCorp's published FY2025 Annual Report to confirm we describe the same workforce. Headcounts matched exactly. The attrition metric did not match its label.

| Department           | AR “voluntary attrition” | Our total attrition | Our voluntary only | Annualised voluntary on active |
|----------------------|--------------------------|---------------------|--------------------|--------------------------------|
| Retail Banking       | 9.3%                     | 9.3%                | 7.7%               | 4.2%                           |
| Technology           | 10.4%                    | 10.4%               | 8.5%               | 4.7%                           |
| Risk & Compliance    | 11.8%                    | 11.8%               | 9.6%               | 5.5%                           |
| Insurance            | 10.3%                    | 10.3%               | 8.0%               | 4.5%                           |
| Wealth Management    | 10.5%                    | 10.5%               | 7.9%               | 4.4%                           |
| Corporate Operations | 11.6%                    | 11.6%               | 9.7%               | 5.5%                           |
| Executive Leadership | 8.3%                     | 8.3%                | 7.4%               | 4.0%                           |
| Group                | 10.4%                    | 10.4%               | 8.5%               | 4.7%                           |

Active employees, departures and all seven department headcounts matched the Annual Report exactly, which is what makes the attrition column meaningful. We restate the metric and flag the definition gap. We do not use it as an accusation.

Seven independent exact matches on the attrition column is not coincidence. The published metric is total attrition, including the 267 involuntary exits.

# Reproducibility

Every figure in this deck regenerates from the four supplied CSVs. No seeds, no sampling, no external benchmarks.

| Script                                   | Produces                                                                                                               |
|------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| ethics_finance/cost_model.py             | Every dollar figure in the deck, the Annual Report reconciliation, the sensitivity grid and the entity attrition table |
| ethics_finance/ethics_audit.py           | Flag consistency, all four-fifths tests, flag precision and recall                                                     |
| ethics_finance/make_figures.py           | The six E-charts, including E6 on slide 5                                                                              |
| explore_part1.py                         | Part 1's 21 charts, including slide 6's entity chart                                                                   |
| explore_pt2/explore_part2.py             | Part 2's 15 charts and 18 findings, including slides 7 and 8                                                           |
| recommendation/cost_fix2.py              | Independent cross-check of the corrected early-tenure lever                                                            |
| recommendation/reconcile_deck_numbers.py | The denominator audit and population table behind A2 and A14                                                           |
| recommendation/make_deck_figures.py      | Slide 9's entity recovery chart                                                                                        |
| recommendation/build_deck.py             | This deck                                                                                                              |

Dependencies are pinned in requirements.txt. From a clean clone: create the venv, install, then run each script from its own directory. Two independent implementations agree on the early-tenure lever; four agree on the entity rates.

# What we cut, and why

*Decided deliberately, not by running out of room. All remain in the repo and are defensible if raised.*

| Finding                                                                              | Why it is not in the deck                                                                                                                                                                               |
|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Recognition gap — under-recognised high performers leave at 9.6% vs 8.1%             | A 1.2× lift is weak next to Entity_B's 1.9×. Doesn't survive a “how big is that really?” question.                                                                                                      |
| Pay equity by level and gender                                                       | Gaps are $\leq 1.0\times$ where samples are large, and only appear at L5–L7 where $n < 80$ . Leading with pay also contradicts slide 7, where Entity_B is paid above NovaCorp-Origin and leaves anyway. |
| “Frozen middle” — attrition flat across L1–L4                                        | A null result. Useful for ruling out a middle-management framing, but no recommendation attaches to it.                                                                                                 |
| Manager Pareto and psychological safety by team                                      | The manager finding is now a negative. It appears on slide 10 as something we rule out, so it does not need its own slide.                                                                              |
| “68% of exits are push, therefore preventable”                                       | Cut on correctness, not space. Push includes 267 involuntary exits; the brief defines it as involuntary-or-managed, not disengagement-driven. See A6 §1.5.                                              |
| “Employees go silent before they quit”                                               | Cut on correctness. There is no pre-exit decay. Leavers respond at 70.6% in wave 1 and 70.3% in wave 5, and the gap to stayers narrows. Reframed on slide 8 as a stable signal available from arrival.  |
| “91% of early exits are acquisition-sourced” and “agency is one of our best sources” | Both retracted. Artifacts of the left-truncation problem in A6 §1.1. On a like-for-like population agency is the second-worst source, not the second-best. See A7.                                      |

# Risk & Compliance, and the department view

NovaCorp's Annual Report singles out Risk & Compliance. We looked, and it is real — but it is not where we would act first.

| Department           | High-value exits | Cost/yr | Roster | High-value loss rate |
|----------------------|------------------|---------|--------|----------------------|
| Corporate Operations | 82               | \$7.7M  | 1,509  | 5.4%                 |
| Risk & Compliance    | 83               | \$8.0M  | 2,022  | 4.1%                 |
| Wealth Management    | 59               | \$5.5M  | 1,549  | 3.8%                 |
| Insurance            | 67               | \$6.1M  | 1,781  | 3.8%                 |
| Retail Banking       | 107              | \$9.3M  | 3,280  | 3.3%                 |
| Technology           | 96               | \$8.0M  | 3,032  | 3.2%                 |
| Executive Leadership | 5                | \$0.5M  | 230    | 2.2%                 |

The Annual Report is right to be concerned. R&C loses high-value people at a materially higher rate than the large divisions. Technology carries 50% more headcount and loses the same dollars.

Corporate Operations is worse still, at 5.4%, and appears nowhere in the Annual Report's commentary. We raise it as a finding requiring follow-up, not a recommendation, because we have not established a mechanism.

Why we didn't lead with R&C. FAR imposes personal liability on senior regulatory staff from March 2024. That is a pull problem in a tight external market, and the classic response is compensation benchmarking, which the Annual Report already commits to. We would be recommending what NovaCorp has already decided to do.

Entity\_A is the proven precedent. It ran the same integration and now sits at 3.6%/yr, below NovaCorp-Origin's own rate. There is no equivalent proof that a compensation response fixes FAR-driven regulatory attrition.

Both cuts are in the data and both are correct. We chose the one with an identified cause, a proven fix and a guided budget already attached.

## Where every number comes from

Each figure quoted on a slide, traced to the script that produces it. Run the script, get the number. Two independent implementations agree on the early-tenure lever; four agree on the entity attrition rates.

| Slide | Figure                                 | Value                               | Source                |
|-------|----------------------------------------|-------------------------------------|-----------------------|
| 3     | Regrettable as counted / value-based   | \$14.3M → \$45.1M                   | cost_model.py \$1     |
| 3     | High performers never counted          | 371                                 | cost_model.py \$1     |
| 5     | Three buckets restated                 | \$45.1M · \$17.8M · \$2.3M          | cost_model.py \$1-3   |
| 5     | Restated total against Finance's \$42M | \$65.2M                             | cost_model.py         |
| 6     | Attrition by legacy entity             | 4.5 / 3.6 / 7.0 / 4.6 %/yr          | cost_model.py \$6     |
| 6     | Entity_B concentration                 | \$8.4M in 1,601 active              | cost_model.py \$5     |
| 7     | Trust and purpose gaps                 | -0.301 and -0.300                   | explore_part2.py \$7g |
| 7     | Compa-ratio                            | 0.961 vs 0.937                      | explore_part2.py \$7f |
| 8     | Entity_B first survey                  | purpose 2.975, trust 3.060          | explore_part2.py \$7i |
| 9     | Entity_A recovery                      | 3.6%/yr, 83.8% response             | make_deck_figures.py  |
| 10    | Senior Manager gap                     | 9.8%/yr vs 2.9%                     | explore_part2.py \$7h |
| 11    | Lever table, all rows                  | \$45.1M / \$17.8M / \$7.9M / \$6.2M | cost_model.py \$5     |
| 12    | Four-fifths ratios                     | 0.08 age · 0.15 level · 0.08 cohort | ethics_audit.py \$3   |
| 12    | Flag precision                         | 263 true / 901 false, 22.6%         | ethics_audit.py \$4   |
| 14    | Annualised voluntary rate              | 4.7%/yr                             | cost_model.py \$0     |
| 14    | Annual Report reconciliation           | 1,400 / 13,403 = 10.4%              | cost_model.py \$0     |

Method notes for the figures restated during the build are in A6 §1.5, §1.7 and A7.



# Does the Entity\_B finding survive controls?

The obvious challenge to slide 6 is that Entity\_B might simply sit in departments or levels that lose people anyway. We tested it. It does not.

| Model                                                                 | Entity_B vs Entity_A | 95% CI      | p                    |
|-----------------------------------------------------------------------|----------------------|-------------|----------------------|
| Controlling for department, role level, pay and high-potential status | 1.95×                | 1.55 – 2.44 | 8.4×10 <sup>-9</sup> |
| ...and additionally for tenure                                        | 1.92×                | 1.53 – 2.41 | 1.6×10 <sup>-8</sup> |

**The effect holds either way.** Slide 6 quotes a 1.9× rate ratio. The adjusted odds ratio is 1.95, so the headline is not an artifact of what Entity\_B does or where it sits.

**Department explains very little.** Only one of six department terms reaches p < 0.05, and every odds ratio sits between 0.77 and 1.00. Risk & Compliance is exactly 1.00 once entity is in the model.

**High-potential status replicates independently.** OR 1.59 (p = 3.6×10<sup>-6</sup>) here, against the 1.64× lift in A8 test 11 computed a different way.

## WHY TENURE IS REPORTED SEPARATELY

tenure\_months is the system-entry artifact from A6 §1.1, so for the acquired cohorts it is close to a restatement of entity itself:

Entity\_C 0-9 months • Entity\_B 0-20 • Entity\_A 7-33 • NovaCorp-Origin 0-462

Controlling for it is close to controlling for the thing being measured, so we report both models rather than picking the flattering one. Baseline is Entity\_A because that is the comparison the deck makes. NovaCorp-Origin is the worst choice of baseline here, since it is the cohort the artifact distorts most.

This does not establish cause. It rules out the most likely confounders, which is a different and weaker claim, and the one we make.

# What to measure, and when you can actually read it

Slide 9 says Entity\_A sets the success measure. This is that measure, stated so it can go in front of a board — and paced to the survey cadence NovaCorp actually runs.

| Metric                        | Entity_B today         | Target (Entity_A today) | First readable | Why this one                                                                   |
|-------------------------------|------------------------|-------------------------|----------------|--------------------------------------------------------------------------------|
| Survey response rate          | 62.6%                  | 83.8%                   | Next wave      | Moves first and costs nothing to measure. The leading indicator.               |
| Senior leadership trust       | 3.051                  | 3.352                   | Next wave      | One of the two dimensions that actually collapsed.                             |
| Purpose & meaning             | 3.058                  | 3.358                   | Next wave      | The other one.                                                                 |
| Employees never surveyed      | 152 on B and C systems | 0                       | Next wave      | You cannot manage a cohort you are not asking. Fixable before the wave issues. |
| Senior Manager (L3) attrition | 9.8%/yr                | 2.9%/yr                 | 2 quarters     | The layer that has to carry the integration message.                           |
| Voluntary attrition           | 7.0%/yr                | 3.6%/yr                 | 3–4 quarters   | The outcome. Moves last, so do not judge the programme on it early.            |

BEFORE THE NEXT WAVE

Redefine the regrettable flag, separate it from the function that approves exits, restate the baseline, and add the 152 unsurveyed staff to the issue list. All policy, all ~\$0.

AT THE NEXT WAVE

The first real read. Response rate and the two collapsed dimensions, against Entity\_A's trajectory rather than Entity\_B's own past. Attrition will not have moved.

THE WAVE AFTER

Confirm direction rather than declare success. Two consecutive wave-on-wave improvements in response rate is the earliest defensible signal.

Anchored to waves, not days: NovaCorp's ran 122, 92, 123 and 181 days apart, mean 4.2 months and widening. A 90-day plan would promise a reading the cadence cannot deliver — and a leading indicator read twice a year is not a leading indicator.

Baselines from cost\_model.py. Targets are Entity\_A's present position.

## Model card — cohort diagnostic

*The accompanying dashboard is a model, so it gets a model card. Intended use, prohibited use, and known limits, stated up front rather than discovered later.*

### INTENDED USE

- Cohort-level diagnostic for HR and integration leads.
- Identifying which parts of the organisation resemble Entity\_B before its attrition rose, so integration effort can be directed.
- Tracking cohorts against the A16 measures over time.

### PROHIBITED USE

- Any decision about an individual. The file contains no employee, manager or team identifier, so this is enforced by construction rather than by policy.
- Performance management, redundancy selection, or promotion input.
- Publishing cohort scores to managers without the fairness context in A4.

### KNOWN LIMITS

- Diagnostic, not predictive. It has no validated forward accuracy and none is claimed.
- Engagement is observed only for responders, so every score is a lower bound on a partially observed population.
- Component weights (40/40/20) are a judgement, not a finding. They are shown separately in the tool so the score can be audited.
- Eight cohorts covering 44 people are suppressed below the response floor.

### Why there is no individual-level version

We built and tested one. On the four-fifths rule it returns impact ratios of 0.08 on age, 0.15 on role level and 0.08 on acquisition cohort, and its precision is 22.6% — 901 false positives against 263 true ones. Low response is concentrated in the acquisition cohorts, so at individual level the flag would substantially be measuring integration failure rather than intent. We did not ship it, and this page exists so that decision travels with the tool.